RamraoAdik Institute of Technology, Nerul, Navi Mumbai

Question Bank-1

Academic year: 2025-2026

Sbject: Embedded System Class: S.E. Sem : III

1. Convert $(73.301)_{10}$ into binary, octal and Hexadecimal equivalent.
2. Convert the following.
   i) $(BC)_{16} = (\ )_{10}$
   ii) $(2314)_8 = (\ )_{10}$
   iii) $(1000011)_2 = (\ )_{10}$
   iv) $(647)_{10} = (\ )_{16}$
3. Express the following numbers in decimal.
   i) $(10110.0101)_2$
   ii) $(16.5)_{16}$
   iii) $(26.24)_8$
4. Give the truth table and symbol of basic logic gates.
5. Give the truth table and symbol of Universal logic gates.
6. Give the truth table and symbol of exclusive logic gates.
7. Design & implement the Half Adder using logic gates.
8. Design & implement the Full Adder using logic gates.
9. What is Multiplexer? Draw a 4:1 Multiplexer, write down the Boolean equation and design it with logic gates.
10. Explain the working of 4:1 Demultiplexer. Design it with logic gates.
11. Draw the circuit of SR Latch and explain its working with truth table.
12. Draw the circuit of D- flip flop and explain its working with truth table.
13. Draw the circuit of SR flip-flop and explain its working with truth table.